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High pressure luminescence of Nd^{3+} in YAlO_3 perovskite nanocrystals: A crystal-field analysis

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Pressure-induced energy blue- and red-shifts of the $^4F_{3/2} \rightarrow ^4I_{9/2,11/2}$ near-infrared emission lines of Nd^{3+} ions in YAlO_3 perovskite nano-particles have been measured from ambient conditions up to 29 GPa. Different positive and negative linear pressure coefficients have been calibrated for the emission lines and related to pressure-induced changes in the interactions between those Nd^{3+} ions and their twelve oxygen ligands at the yttrium site. Potentiality of the simple overlap model, combined with *ab initio* structural calculations, in the description of the effects of these interactions on the energy levels and luminescence properties of the optically active Nd^{3+} ion is emphasized. Simulations show how the energies of the $4f^3$ ground configuration and the barycenters of the multiplets increase with pressure, whereas the Coulomb interaction between f -electrons decreases and the crystal-field strength increases. All these effects combined explain the wavelength blue-shifts of some near-infrared emission lines of Nd^{3+} ions. Large pressure rates of various emission lines suggest that a YAlO_3 perovskite nano-crystal can be a potential candidate for near-infrared optical pressure sensors. *Published by AIP Publishing.* <https://doi.org/10.1063/1.5010150>

I. INTRODUCTION

High-pressure research involves different science fields, such as physics, chemistry, geology, biology, and food technology.¹ Applying pressure to an organic or inorganic material allows tuning, in a controllable way, the volume of the sample, changing their interatomic distances and structural distribution. Therefore, this process modifies, in a reversible or an irreversible way, its structural, vibrational, electric, magnetic, and optical properties. Among several applications of this technique, such as reproducing and studying the phenomena that occur in the inner cores of planetary objects, it is really interesting to highlight its application in the study of structural and optical properties of inorganic compounds since it presents several advantages over ambient pressure condition techniques.^{2,3} On the one hand, it provides a continuous change of the network structure and the local coordination environments, allowing a singular view of the electronic structure and optical properties of materials doped with lanthanide ions and, hence, of all their physical properties. On the other hand, reduction of the sample's volume allows a wide range of structures and bindings of a specific chemical composition to be achieved.^{2,3}

Since its development in the 1960s, a diamond anvil cell (DAC) has been the usual way to reach ultrahigh pressures.¹ Its use requires calibrated standards for the determination of

pressure inside the hydrostatic chamber. The most widely used sensors are the optical pressure ones (P-sensors), taking advantage of variations of the luminescence properties of optically active ions under pressure. An optical P-sensor needs to accomplish some general requirements, such as a single emission line or well separated lines with no broadening or weakening, large shifts with pressure, and small temperature-dependent line shifts and linewidths.⁴ The $\alpha\text{-Al}_2\text{O}_3\text{:Cr}^{3+}$ ruby crystal^{1-3,5} is the most commonly used optical P-sensor due to the intense $^2E \rightarrow ^4A_2$ (R-lines) luminescence of Cr^{3+} ions, their large line shifts with pressure, and easy excitation with commercial lasers (argon or diodes). However, its main drawbacks are the broadening and loss of intensity at high temperatures as well as a low-pressure sensitivity below 1 GPa, the pressure range of interest for life-related high-pressure research.^{2,5} Therefore, the development of new optical P-sensors is required in order to overcome these disadvantages while keeping the advantages of ruby. For this purpose, the lanthanide-based sensors have drawn special attention.^{2,3} Due to the shielding of $4f$ electrons, absorption and emission lines of lanthanides in crystals in the ultraviolet-visible-near infrared (UV-VIS-NIR) optical range are quite sharp and, consequently, pressure measurements result to be more accurate. In addition, these lines are less sensitive to the environment and temperature compared to those of $3d$ electrons of Cr^{3+} ions in ruby.

Several oxide-based active materials have been extensively studied as optical P-sensors.^{2,3} Among them,

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neodymium(III)-doped garnets and perovskites have received special attention due to their optical properties, since the main near-infrared (NIR) emission in the range of 850–1100 nm may avoid the luminescence of impurities present in natural diamonds.^{6–14} Different Nd^{3+} -doped or Cr^{3+} - Nd^{3+} co-doped systems such as $\text{Ca}_3\text{Ga}_2\text{Ge}_3\text{O}_{12}$ (CGGG),¹⁵ $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ (GGG),¹⁶ $\text{Y}_3\text{Al}_5\text{O}_{12}$ (YAG),¹⁷ $\text{La}_3\text{Lu}_2\text{Ga}_3\text{O}_{12}$ (LLGG),¹⁸ and $\text{Gd}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ (GSGG)¹⁹ garnets have been presented as potential P-sensors, focusing on the pressure-induced red-shifts of the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{9/2}(\text{Z}_5)$ emission lines of the Nd^{3+} ion (see Fig. 1). In addition, a Nd^{3+} -doped YAlO_3 (YAP) perovskite has been also studied because of its small temperature coefficient at high temperatures, seven times lower than ruby,⁴ and by the fact that its emission is located near 900 nm, just in the NIR, which does not overlap with typical emissions in the visible range from other optically active ions. Luminescence of Nd^{3+} ions in the YAP bulk crystal has been previously tested up to 80 GPa measuring the pressure-induced blue-shifts of the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{9/2}(\text{Z}_{1-3})$ lines (see Fig. 1),⁶ but nothing was said about the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{9/2}(\text{Z}_{4,5})$ lines that are the most sensitive to pressure, as well as the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{11/2}(\text{Y}_{1-6})$ lines.

From the luminescence point of view, advantages of nanocrystals versus bulk counterparts are found, especially in the field of nano-bioimaging and nano-thermometry and as alternative to quantum dots in photonic devices.^{20–23} In our case, luminescence properties of nanocrystals of sizes around 30 nm seem to show the same behavior of their bulk counterparts. In this work, and for the first time to our knowledge, high-pressure luminescence properties of $\text{YAP}:\text{Nd}^{3+}$ have been extended to nanocrystals, focusing the attention not only on the emission from the $^4\text{F}_{3/2}$ multiplet to the $^4\text{I}_{9/2}$ ground state but also to the $^4\text{I}_{11/2}$ first excited multiplet, in the pressure range from 1 atm up to 29 GPa. High-pressure

behavior indicates different wavelength blue- and red-shifts of the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{9/2}(\text{Z}_{1-5})$ and $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{11/2}(\text{Y}_{1-6})$ emission lines, showing pressure coefficients large enough to consider the use of this Nd^{3+} -doped nanocrystal as an optical P-sensor in the NIR range. These pressure-induced wavelength shifts are analyzed and simulated in the framework of the crystal-field (CF) theory, taking advantage of the combination of the Simple Overlap Model (SOM) and *ab initio* structural calculations.

II. EXPERIMENTAL

Crystalline YAP perovskites in the form of nano-powder with composition $\text{Y}_{0.99}\text{Nd}_{0.01}\text{AlO}_3$ were successfully synthesized by the Pechini citrate sol-gel method in air atmosphere. Details of the synthesis procedure are given elsewhere¹⁴ although it is worth noting that a high temperature of 1550 °C (1823 K) was reached in the final calcination process in order to obtain a pure perovskite phase. High pressure was generated with a membrane DAC from ambient pressure up to ~29 GPa at room temperature. A 200 μm tungsten carbide gasket was pre-indented to 50 μm , and a centered hole was made using a spark eroder to make a hydrostatic chamber with a diameter of around 150 μm . Luminescence of ruby sphere chips of around 10 μm was used as a pressure calibrant, and a 16:3:1 methanol-ethanol-water mixture served as the pressure transmitting medium, providing hydrostatic conditions up to ~14 GPa.³ High-pressure luminescence of Nd^{3+} ions was detected using a micro-Raman confocal microscope (Renishaw InVia), using a 785 nm laser and a cooled CCD.

Powder XRD data at ambient conditions were collected on a diffractometer (PANalytical X'Pert PRO) in a Bragg-Brentano geometry with a detector (X'Celerator) using the $\text{Cu K}\alpha_1$ radiation ($\lambda = 1.5405 \text{ \AA}$) in the angular range

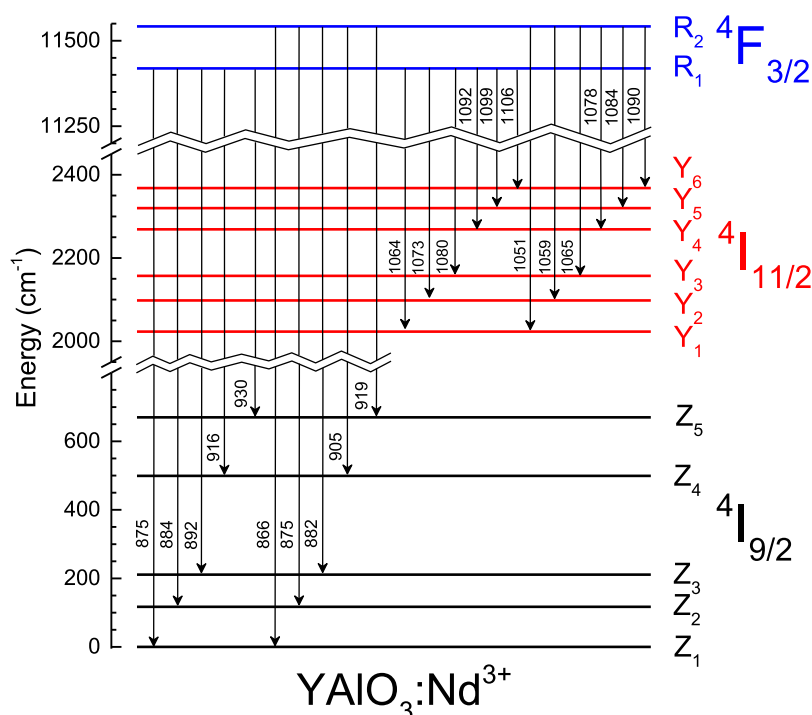


FIG. 1. Partial energy level diagram and main emission lines of Nd^{3+} ions in YAlO_3 nano-perovskite at ambient pressure. Numbers in arrows are given in nanometers.

$5^\circ < 2\theta < 80^\circ$, by continuous scanning with a step size of 0.02° . A Rietveld refinement²⁴ of the pattern was carried out using the FullProf program,²⁵ in which a pseudo-Voigt function was used to describe the peak shape and a polynomial function with five refined coefficients for the background. Table T1 included in [supplementary material](#) shows the cell parameters and agreement factors obtained from Rietveld fitting. Results are the same as those presented previously.^{14,26} Structural properties of the local dodecahedral site occupied by the yttrium(III) and neodymium(III) ions at ambient as well as at high pressures were theoretically calculated for bulk YAlO_3 perovskite by means of *ab initio* total energy calculations using the Density Functional Theory (DFT).²⁶ Crystal-field definitions and Simple Overlap Model (SOM) simulation details are described in [supplementary material](#).

III. RESULTS AND DISCUSSION

A. Crystal structure

The diffraction pattern of the YAP:Nd^{3+} (1%) nano-powder sample has been well indexed to a major orthorhombic lattice with space group $Pnma$,¹⁴ with an average grain size of around 35 nm, determined from the Scherrer formula. No amorphous phase was found in it, and the crystal structure parameters ($a = 5.33 \text{ \AA}$, $b = 7.38 \text{ \AA}$, $c = 5.18 \text{ \AA}$) obtained for this nano-powder after fitting are almost identical to those found for the bulk counterpart ($a = 5.33 \text{ \AA}$, $b = 7.37 \text{ \AA}$, $c = 5.18 \text{ \AA}$),²⁶ indicating that the structural properties of the bulk remain in the nanoscale.

Ab initio calculation of the bulk modulus for the YAlO_3 structure gives a large value of 204.5 GPa, explaining its hardness and mechanical stability up to 92 GPa, above which a possible phase transition from the $Pnma$ phase to the tetragonal $I4/mcm$ phase may occur.²⁶ Pressure dependence of the YO_{12} dodecahedron (V_{dod}), AlO_6 octahedron (V_{oct}), and unit cell (V_{tot}) volumes is given in Fig. 2 (bottom). As a general feature, all volumes decrease with pressure although the octahedron volume shows a higher compressibility ($5.6 \times 10^{-3} \text{ GPa}^{-1}$) than the dodecahedron one ($4.35 \times 10^{-3} \text{ GPa}^{-1}$), indicating an overall pressure evolution of the YAlO_3 network towards a more symmetrical structure.²⁶ This can also be observed when following the evolutions of the interatomic distances between the Y^{3+} , or Nd^{3+} , ion and its oxygen ligands (see the top panel of Fig. 2). At ambient pressure, the coordination number of the Y^{3+} , or Nd^{3+} , ion in YAlO_3 is practically smaller than 12 (around $8 + 4$) since there are 4 oxygens at distances above 3 Å. In addition, these 8 closer oxygens show small changes in their distances to the yttrium ion with pressure. However, the large compressibility of the distances of the other 4 outer oxygens at high pressures clearly indicates that they start to have more influence on the environment of the Nd^{3+} site. That is why SOM calculations have used all 12 oxygen ligands in all the calculations of the crystal-field parameters. On the other side, it should be emphasized that while the studied (doped or un-doped) YAP phase is orthorhombic, the stoichiometric NdAlO_3 (NdAP) compound is rhombohedral,²⁷ with different distances between the Y^{3+} , or Nd^{3+} , ions and the oxygens occupying the dodecahedral environments in both structures.

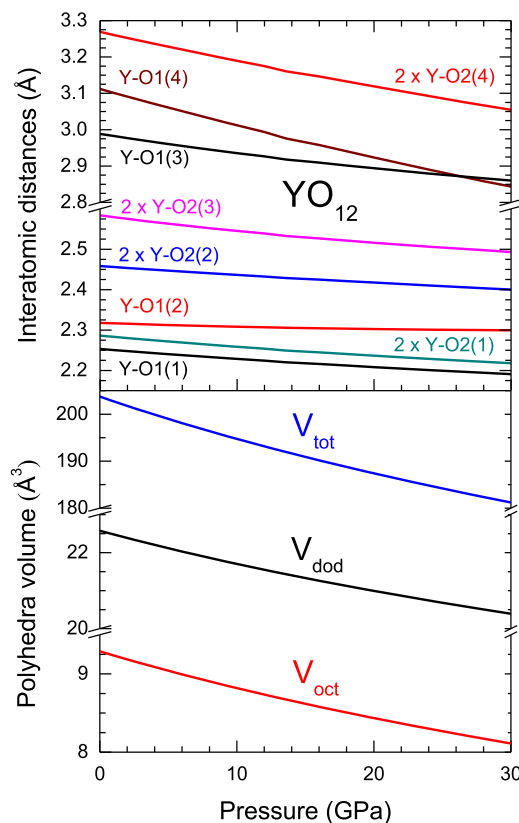


FIG. 2. Pressure evolution up to 30 GPa of the *ab initio* calculated (top) yttrium-oxygen interatomic distances in the YO_{12} dodecahedral site and (bottom) of the unit cell, dodecahedral, and octahedral volumes in the bulk YAlO_3 structure.

Thus, SOM calculations have been carried out assuming that Nd^{3+} ions occupy the yttrium lattice site without any distortion although, since the Nd^{3+} ion ($0.98 \text{ \AA}$) is about 10% larger than the Y^{3+} ion ($0.90 \text{ \AA}$), local distortions are expected. Unfortunately we cannot predict what kind of distortion can be introduced when the Nd^{3+} ion is located in the Y^{3+} site (it could be strong since NdAlO_3 has a different structure than YAlO_3). In other words, it is not clear how this deformation can be quantified and introduced in the calculation. In spite of this fact, it seems evident that the simulation carried out properly reproduces the observed behavior with the pressure of the energy levels. Even, in some cases, the position of the emission lines with the pressure is properly reproduced, especially in the high-energy side of each transition and particularly for the intense $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{9/2}(\text{Z}_{1,25})$ transitions that, taking place at 860-900 nm, are easily measured using most of the conventional fluorimeters.

B. Luminescence under pressure

Luminescence properties of Nd^{3+} ions, as the rest of optically active lanthanides, are ruled by the local point symmetry of the site, or local environment, of the lattice in which it is located. The point symmetry group at the yttrium dodecahedral YO_{12} site is C_S (C_{1h}), which includes a reflection plane in addition to the identity operation. Due to the low point symmetry of the Nd^{3+} sites in the YAlO_3 structure, probably even slightly lower due to the distortion expected for an optically active ion with larger ionic radius compared to that of the Y^{3+}

ion in the perovskite network, there is a complete breakdown of the $(2J + 1)/2$ degeneracies of the $^{2S+1}L_J$ multiplets. Thus the C_S (C_{1h}) crystal-field interaction will give rise to the R_1 and R_2 Stark, or crystal-field, levels of the $^4F_{3/2}$ multiplet, the Y_j ($j = 1-6$) Stark levels of the $^4I_{11/2}$ first excited multiplet, and the Z_i ($i = 1-5$) levels of the $^4I_{9/2}$ ground state, the ones that are relevant in this work. Transitions between these Stark levels will show the largest luminescence intensities (see Figs. 1 and 3).^{15,28}

The emission spectrum at ambient conditions has been measured resonantly exciting at 785 nm the $^4I_{9/2} \rightarrow ^4F_{5/2}$ absorption transition of the Nd^{3+} ions in YAP nanocrystals,

from which a fast multiphonon relaxation to the $^4F_{3/2}$ doublet occurs. Spontaneous emissions from the R_1 and R_2 thermalized Stark levels of the $^4F_{3/2}$ emitting level to those of $^4I_{9/2}$, in the range from 860 to 940 nm, and to the $^4I_{11/2}$ Stark levels, from 1040 to 1110 nm, result in well-defined narrow peaks that can be assigned to nine $^4F_{3/2}(R_{1,2}) \rightarrow ^4I_{9/2}(Z_{1-5})$ transitions and twelve $^4F_{3/2}(R_{1,2}) \rightarrow ^4I_{11/2}(Y_{1-6})$ transitions, respectively (see Fig. 3). From these spectra, an identification of the energy level scheme of the Stark levels and the emission lines relevant for this work at ambient pressure are shown in Figs. 1, 3, and 4. The range of hydrostaticity inside the high-pressure chamber was followed monitoring the widths of the ruby emission

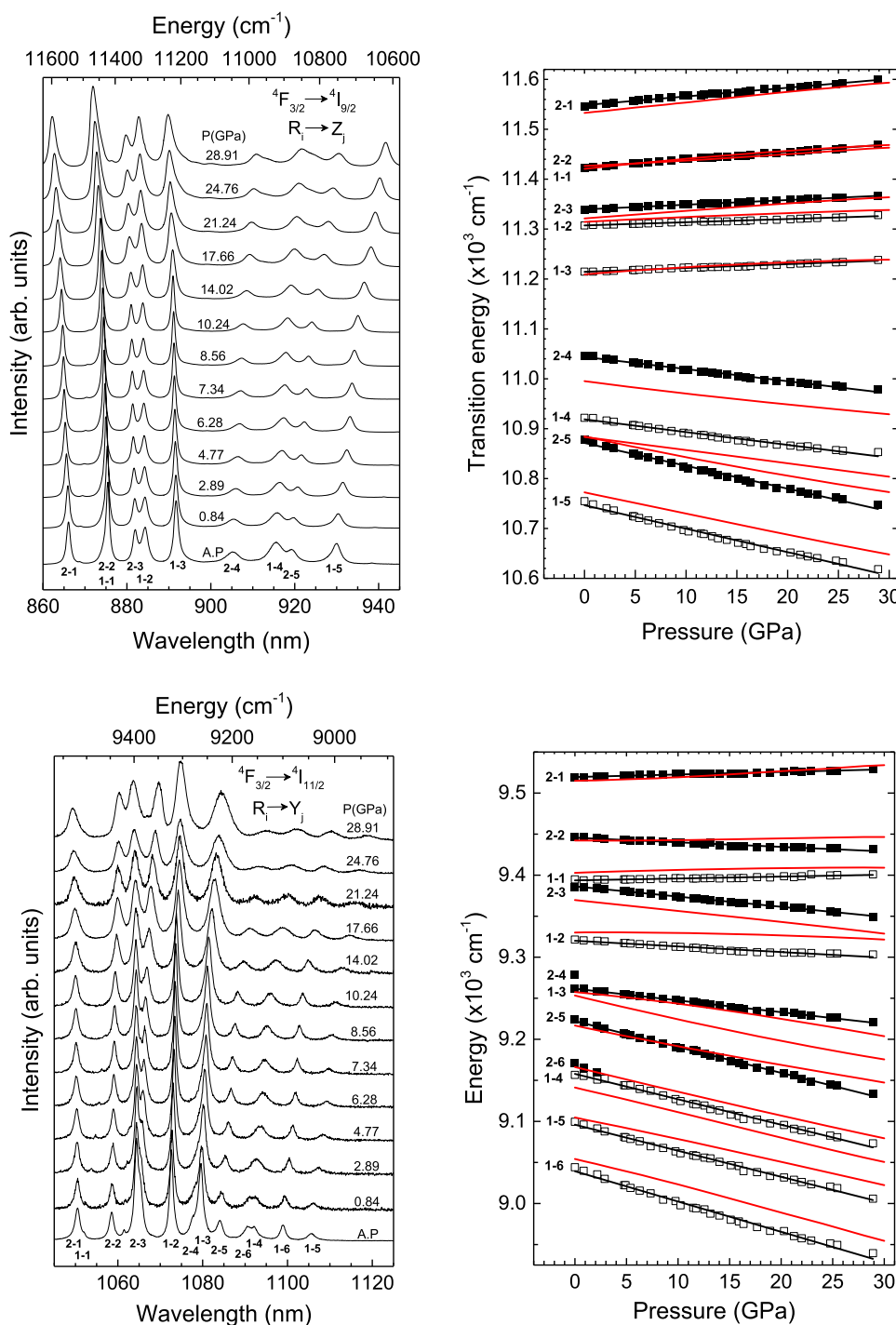


FIG. 3. (Left) Room temperature Nd^{3+} emission spectra as a function of pressure associated with the $^4F_{3/2}(R_{1,2}) \rightarrow ^4I_{9/2}(Z_{1-5})$ and $^4F_{3/2}(R_{1,2}) \rightarrow ^4I_{11/2}(Y_{1-6})$ transitions exciting resonantly at 785 nm the $^4I_{9/2} \rightarrow ^4F_{5/2}$ transition in YAlO_3 nano-perovskites. Identifications of the $R_i \rightarrow Z_j$ and $R_i \rightarrow Y_j$ transitions between Stark levels are also given. (Right) Experimental (open symbols) and linearly fitted (black lines) energies of emission lines as a function of pressure. Crystal-field simulations of the emission energies (red lines) using the Simple Overlap Model (SOM) are also shown.

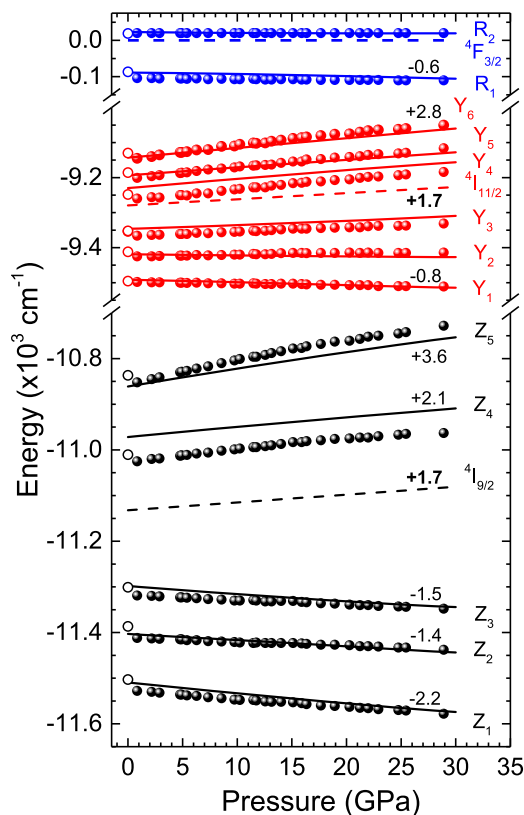


FIG. 4. Nd^{3+} Stark energy levels under pressure in YAlO_3 nano-perovskite obtained from the emission lines (solid symbols) and from the simulations using the Simple Overlap Model (SOM) (solid lines). Experimental Stark levels by Duan *et al.*³⁰ at ambient conditions (open symbols) are also given. Barycenter of the $^4\text{F}_{3/2}$ emitting level (blue dashed line) has been taken as the zero of energy and some relative energy shift rate values of simulated Stark levels and barycenters (dashed lines) are highlighted.

lines. In our case, measurements revealed that up to 16 GPa it is maintained, a fact also reflected in the emissions bands of the Nd^{3+} ions, which remain wide and sharp during this pressure range, with only small broadenings observed at pressures above 16 GPa in the wavelength range of 1080–1110 nm.

The main feature in these spectra is that two different behaviors can be observed in the evolution of the energies of the emission lines with pressure, particularly for transitions between Stark levels of the $^4\text{F}_{3/2}$ emitting level and the $^4\text{I}_{9/2}$ ground state: On the one side, the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{9/2}(\text{Z}_{1,2,3})$ emission lines show wavelength blue-shifts (to lower wavelengths—higher energies) as the volume of the nanocrystal decreases, in agreement with those obtained in its bulk counterpart,⁶ especially the $\text{R}_2 \rightarrow \text{Z}_1$ transition for which this shift is more evident. On the other side, the most sensitive transitions to pressure, the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{9/2}(\text{Z}_{4,5})$ emission lines, show significant wavelength red-shifts, as it has been also observed in garnet crystals.^{15–19} Regarding the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{11/2}(\text{Y}_{1-6})$ transitions, emission lines associated with the $\text{R}_{1,2} \rightarrow \text{Y}_{2-6}$ emission lines show clearly wavelength red-shifts, whereas only the $\text{R}_2 \rightarrow \text{Y}_1$ line shows a slight blue-shift (see Fig. 3). As it will be shown later, these behaviors are related to different energy shift rates of the Stark levels with pressure, which unexpectedly make the emissions from the $\text{R}_{1,2}$ Stark levels to the Z_{1-3} ones to shift to lower wavelengths (higher energies).

In the range from ambient pressure up to ~ 29 GPa, evolution of the emission lines and, hence, of the Stark energy levels can be described by simple linear equations although at higher pressures the fitting should include higher order polynomials.⁶ Thus for high-pressure calibration purposes, in a first approximation, linear pressure evolutions of the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{9/2}(\text{Z}_{1-5})$ and $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{11/2}(\text{Y}_{1-6})$ emission lines were considered in order to obtain a scale in the range of pressures achieved (see Fig. 3 and Table T2; the latter is included in [supplementary material](#)). The $\text{R}_{1,2} \rightarrow \text{Z}_5$ emission lines resulted to be the most sensitive to pressure with coefficients of -4.70 and $-4.58 \text{ cm}^{-1}/\text{GPa}$, respectively. Concerning the spectral range of the $^4\text{F}_{3/2} \rightarrow ^4\text{I}_{11/2}$ transition, it is worth noting that the $\text{R}_1 \rightarrow \text{Y}_{4,5,6}$ lines show pressure coefficients around -3.10 , -3.18 , and $-3.69 \text{ cm}^{-1}/\text{GPa}$, respectively, and the $\text{R}_2 \rightarrow \text{Y}_5$ one with $-3.15 \text{ cm}^{-1}/\text{GPa}$. These rates are smaller than ruby emission lines ($-7.35 \text{ cm}^{-1}/\text{GPa}$),⁵ but they are good enough to consider the use of YAP:Nd^{3+} nano-perovskite as an optical P-sensor in the NIR range.

Since the YAlO_3 crystal has been one of the most successful lanthanide laser host materials, just behind the $\text{Y}_3\text{Al}_5\text{O}_{12}$ (YAG) garnet, its energy level diagram is well known up to the $^2\text{P}_{1/2}$ level at around $23\,160 \text{ cm}^{-1}$ ⁷ and to the $^2\text{P}_{3/2}$ doublet located at $25\,981$ and $26\,123 \text{ cm}^{-1}$ ²⁹ (see Table T3 in [supplementary material](#)) although recent studies increase the number of Stark levels into the UV range.³⁰ In addition, from emission spectra, it has been possible to extract the energy level scheme of those multiplets relevant for this work, i.e., the $^4\text{I}_{9/2}$ ground, $^4\text{I}_{11/2}$ first excited, and $^4\text{F}_{3/2}$ lowest emitting levels of the Nd^{3+} ions in YAlO_3 nano-perovskite at ambient conditions, which are compared to those of Duan *et al.*,³⁰ as well as at high pressures (see Fig. 4).

A rough estimation of the maximum splitting increasing rates of these multiplets up to around ~ 29 GPa, and assuming linear variations, gives rise to $\Delta E(\text{Z}_5\text{-Z}_1) = +6.2 \text{ cm}^{-1}/\text{GPa}$ for the $^4\text{I}_{9/2}$ ground multiplet, $\Delta E(\text{Y}_6\text{-Y}_1) = +3.85 \text{ cm}^{-1}/\text{GPa}$ for the $^4\text{I}_{11/2}$ multiplet, and $\Delta E(\text{R}_2\text{-R}_1) = +0.22 \text{ cm}^{-1}/\text{GPa}$ for the $^4\text{F}_{3/2}$ emitting multiplet. These results clearly point to a pressure-induced increase of the crystal-field strength felt by the Nd^{3+} ion when distances to its oxygen ligands decrease, as it is usually observed for many systems.^{2,3} The extremely low rate of the splitting $\Delta E(\text{R}_2\text{-R}_1)$ of the $^4\text{F}_{3/2}$ level compared to that of the ground and first excited multiplets would play a relevant role in the explanation of the wavelength blue-shifts of many emission lines associated with transitions between Stark levels of these multiplets, i.e., the $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{9/2}(\text{Z}_{1-3})$ and $^4\text{F}_{3/2}(\text{R}_{1,2}) \rightarrow ^4\text{I}_{11/2}(\text{Y}_1)$ emission lines.

A simple comparison with other crystalline matrices let us say that the rate of $+6.2 \text{ cm}^{-1}/\text{GPa}$ for the maximum splitting $\Delta E(\text{Z}_5\text{-Z}_1)$ of the $^4\text{I}_{9/2}$ ground multiplet is smaller than $+10 \text{ cm}^{-1}/\text{GPa}$ found in CGGG¹⁵ or $+8.1 \text{ cm}^{-1}/\text{GPa}$ in GSGG garnets.¹⁹ However, it is worth noting that a completely different behavior is observed for the rate of change of the maximum splitting $\Delta E(\text{R}_2\text{-R}_1)$ of the $^4\text{F}_{3/2}$ multiplet compared to what is found in garnets, which decreases with negative rates of $-5.0 \text{ cm}^{-1}/\text{GPa}$ in YAG, $-2.6 \text{ cm}^{-1}/\text{GPa}$ in GSGG, or $-2.0 \text{ cm}^{-1}/\text{GPa}$ in CGGG¹⁵ due to structural atomic redistribution with pressure, which effectively reduces

the orthorhombic distortion making the site more “cubic,” i.e., with a lower splitting. This splitting has been considered as a measure of the degree of local distortion of the local structure,¹⁵ and since in garnets it is still larger in absolute magnitude than for the perovskite structure, it can be concluded that the local structure around the Nd³⁺ optically active ion does not suffer large distortions in the atomic arrangement with pressure up to ~29 GPa, confirming the *ab initio* calculations.

C. Crystal-field analysis

When a lanthanide ion enters in a solid structure, free-ion $^{2S+1}L_J$ multiplets are red-shifted while their degeneracies are partially or totally lifted by the crystal-field interaction. If pressure is applied to this solid, there is a continuous decrease of its volume, which can be directly understood as decreasing distances between the lanthanide ions and their ligands, together with changes in the relative spatial distributions of those atoms. As pointed out by Tröster,² in almost all cases, absorption and luminescence lines of lanthanide ions shift to lower energies under pressure, an effect that can be understood as an overall contraction of the $4f^N$ ground configuration, which reduces the energy differences between ground and excited multiplets. This effect can be related to decreasing Coulomb and spin-orbit interactions through changes in the covalency of bonds, named the nephelauxetic effect. In addition, a generalization of the crystal-field interaction under pressure can also be given, in such a way that, superimposed on the reduction of the energy gaps between multiplets, there is an increase in their hyperfine Stark energy level splittings due to increasing lanthanide-ligands mutual interaction when distances decrease. However, in some cases also decreasing splittings have been reported, related to specific changes of the structural distribution of the atoms of the lanthanide's local site that decreases the crystal-field strength felt by the optically active ion.^{15–19,31} It is worth noting that the YAlO₃ crystal has been claimed to be one of the scarce exceptions to these generalizations since blue-shifts of some emission peaks have been observed under pressure.^{4,6}

Up to now, a clear explanation of this anomalous behavior in Nd³⁺-doped YAlO₃ has not been given. Some authors suggested as main factors the site symmetry and the shorter distance between the lanthanide and its ligands, showing a higher hybridization of the $4f$ wavefunctions of the Nd³⁺ ion in this system. Another simple explanation would point to increasing Coulomb and spin-orbit coupling interactions, which should be connected with an increasing volume of the YO₁₂ dodecahedral unit,² an effect that does not occur in the YAlO₃ perovskite structure (see Fig. 2).

It is worth noting that the complete picture of the effects of pressure on this system has not been given. In previous studies,^{4,6} there were no measurements of the rest of R_{1,2} → Z_{4,5} emission lines above 900 nm (11 111 cm^{−1}), which show red-shifts with rates even larger than those of blue-shifted lines, as well as the R_{1,2} → Y_{1–6} emissions beyond 10 000 cm^{−1}, with similar results (see Fig. 3). In addition, no attempt to understand how the atomic and crystal-field interactions change with pressure and their influence on the energy levels and emission lines was made.

1. Simulation at ambient pressure

The first attempt to simulate the crystal-field interaction of Nd³⁺ ions in YAlO₃ perovskite was done by Karayianis *et al.*³² by diagonalization of six low- J lying multiplets and using 27 experimental levels. A few more simulations were later published^{33,34} although no correlation exists among the different sets of calculated parameters, mainly due to differences in calculations as well as in the definition of the Hamiltonian used. As a general feature, there is a lack of sufficient experimental energy levels and ambiguities in the identification of high-energy multiplets, due to overlapping, that have complicated these simulations. In addition, small errors in the computer files of reduced matrix elements used in these calculations, recently detected by several authors,^{35–37} introduced additional uncertainty in the reliability of the early calculations. Thus the first reliable calculation was done in 2007 by Duan *et al.*,³⁰ when they revisited the study of the crystal-field interaction felt by Nd³⁺ and Er³⁺ ions in YAlO₃ perovskite.

For a proper parameterization of the crystal-field interaction, most of the energy levels in the ⁴I_J and ⁴F_J multiplets (<15 000 cm^{−1}) are needed. Since in this work the energy level positions obtained from experimental data are clearly not enough for a complete description of the system, the starting point of our analysis using the SOM model is based not only on previous spectroscopic data but also, and this is one of the key points of this work, on the use of crystallographic data of the yttrium site in the perovskite structure.

The first step in our simulation procedure is obtaining the barycenters and the Stark energy levels of all the $^{2S+1}L_J$ multiplets up to the ²P_{1/2} singlet at ~23 160 cm^{−1} (see Table T3 in [supplementary material](#)). In a second step, keeping the same free-ion parameters obtained by Duan *et al.*,³⁰ the SOM model has been used to obtain the crystal-field (CF) parameters (see Table I and simulation II in Table T3 in [supplementary material](#)). These values give an overall CF strength at ambient conditions of 560.9 cm^{−1}. Simulation II gives an uncertainty of around 18 cm^{−1} when compared with the experimental Stark energy levels given by Kaminskii.²⁹ CF parameters in simulations I and II (Table T3) are clearly different although this is not surprising since a local distortion at the optically active ion site is expected when the Nd³⁺ ion is placed at the Y³⁺ lattice position. Using the free-ion of Duan *et al.*³⁰ and the CF parameters of SOM as a starting point, a new fit to the Kaminskii data, named simulation III, is also reported in Table I and Table T3 ([supplementary material](#)), giving a *rms* uncertainty of around 12 cm^{−1}.

2. Simulation at high pressures

To theoretically simulate the pressure evolution of the Nd³⁺ energy levels and, hence, their emission lines involved in these transitions, the total Hamiltonian within the $4f^3$ ground configuration must be solved. The starting point begins with considering the accessible experimental energy transitions between Stark levels obtained from emission spectra under pressure: in our case, only 21 lines correspond to the ⁴F_{3/2}(R_{1,2}) → ⁴I_{9/2}(Z_{1–5}) and ⁴F_{3/2}(R_{1,2}) → ⁴I_{11/2}(Y_{1–6}) transitions. These energies have been used to generate only

TABLE I. Atomic and crystal-field parameters (in cm^{-1}) calculated for the Nd^{3+} ion in the YAlO_3 nano-crystal.^a

Atomic and crystal-field parameters	Simulation I ²³ (0 GPa)	Simulation II (0 GPa)	Simulation III (0 GPa)	This work (30 GPa)
E_{AV}	24 114	24 116	24 116	24 133
$F^{(2)}$	70 925	70 925	70 925	70 565
B_0^2	-154	-487	-160	-319
B_2^2	578	526	604	669
B_0^4	-541	-420	-410	-566
B_2^4	967	594	925	617
B_4^4	24	349	164	607
B_4^6	-309	-484	-332	-732
B_6^6	608	727	599	183
B_0^6	-671	-753	-738	-974
B_2^6	512	151	525	315
B_4^6	-18	-214	2	-193
B_6^6	1 611	1 583	1 586	1 390
S	565.4	560.9	562.8	676.4
$S^{(2)}$	214.8	229.5	224.4	257.9
$S^{(4)}$	338.5	313.3	326.0	330.9
$S^{(6)}$	398.7	404.7	400.1	530.6
$\%S^{(2)}$	14.4	16.7	15.9	14.5
$\%S^{(4)}$	35.8	31.2	33.6	23.9
$\%S^{(6)}$	49.7	52.1	50.5	61.5

^aFixed atomic parameter (in cm^{-1}) from Ref. 23: $F^{(4)} = 50\,794$; $F^{(64)} = 35\,424$; $\zeta_{4f} = 875$; $\alpha = 23$; $\beta = -691$; $\gamma = 1690$; $T^2 = 458$; $T^3 = 38.4$; $T^4 = 75.8$; $T^6 = -290$; $T^7 = 237$; $T^8 = 469$; $M^0 = 1.9$; $M^2 = 1.06$; $M^4 = 0.72$; $P^2 = 206$; $P^4 = 154.5$; $P^6 = 103$.

13 experimental Stark energy levels at different pressures (see Fig. 4). The insufficient experimental data does not allow us to follow any fitting procedure to obtain both atomic and CF parameters using the total Hamiltonian [see Eq. (1) in [supplementary material](#)]. That is why we have chosen an approach similar to that used at ambient conditions: simulate the Stark energy level diagrams using as CF parameters those obtained from the SOM combined with the information of the local structure of the Nd^{3+} first coordination shell at different pressures up to ~ 30 GPa given by *ab initio* calculations. Free-ion parameters obtained in former simulations at ambient pressure were also used but, and this is an important point, assuming slight changes in their magnitudes with pressure.

It is worth noting the importance of the free-ion Hamiltonian in this study since it contributes to the observed shift in the energy level positions due to the pressure-induced displacements of the barycenters of all multiplets. To simulate these displacements, only the $F^{(2)}$ Slater parameter was allowed to vary freely together with the average energy E_{AVG} of the $4f^3$ configuration (see [supplementary material](#)).

Focusing our attention first on the atomic parameters corresponding to the Nd^{3+} free-ion Hamiltonian, depicted in Fig. 5, it can be observed that the E_{AVG} parameter shows a continuous increase with pressure. Up to ~ 20 GPa, an apparent linear behavior is obtained with a slope of $+0.75 \text{ cm}^{-1}/\text{GPa}$, whereas at higher pressures, this rate of growth is gradually

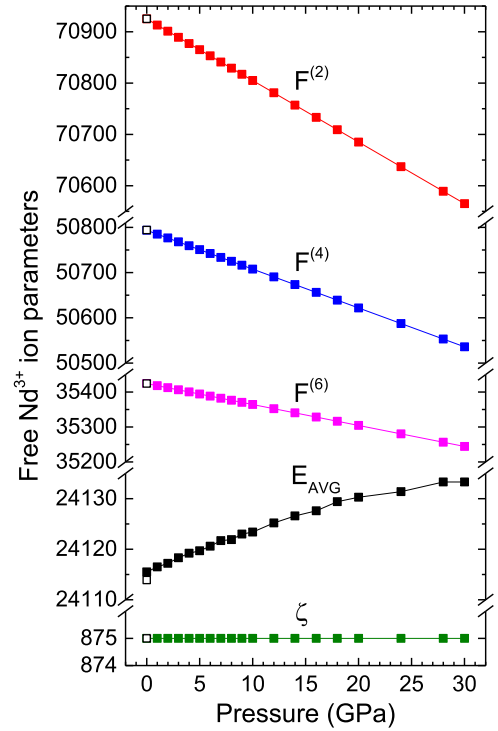


FIG. 5. Nd^{3+} atomic parameters simulated by the Simple Overlap Model (SOM) in YAlO_3 nano-perovskite under pressure. Calculated values obtained by Duan *et al.*³⁰ at ambient conditions (open symbols) are also given.

reduced becoming less linear. The E_{AVG} parameter corresponds to the totally spherically symmetric parts of the total (free-ion plus crystal-field) Hamiltonian, i.e., closed shells of Nd^{3+} ions and oxygen ligands plus their nuclei, giving rise to a repulsive central potential that increases in magnitude with pressure and shifts the whole $4f^3$ configuration to higher energies. On the other side, for the electrostatic repulsion radial integral $F^{(2)}$, a linear behavior is obtained with a slope of $-12.0 \text{ cm}^{-1}/\text{GPa}$, following the expected tendency of the free-ion interaction under pressure. This decrease is due to the increase of covalency effects related to orbital overlapping, giving a relative reduction of $\Delta F^{(2)}/F_{AMB}^{(2)} = -0.51\%$, in the pressure range from ambient up to 30 GPa, similar to those obtained in other systems.² Overall, the final effect, apart from the crystal-field effect, is a blue-shift of the whole $4f^3$ configuration together with a contraction of the $2S+1L$ terms with pressure up to around 30 GPa.

Calculated CF parameters show, as usual in low local symmetries, different variations with pressure (see Fig. 6). The axial parameters of 4- and 6-rank, B_0^4 and B_0^6 , increase in magnitude while 2-rank axial B_0^2 parameter decreases, reflecting that the charge distribution over long distances becomes less axial as pressure brings the distant oxygen ions towards the position of the Nd^{3+} ion in the dodecahedral site (see Fig. 2). Consequently, the distorted environment of the Nd^{3+} ion, formed by 8 nearby oxygens and 4 more distant ones, becomes more symmetric, upholding the *ab initio* calculations.²⁶ As shown in Fig. 2, distances to the latter oxygens change very sharply with pressure. In addition, it is worth noting that the evolution of the interatomic distances, although important, is not the only source of variations in the CF parameters, which also show a

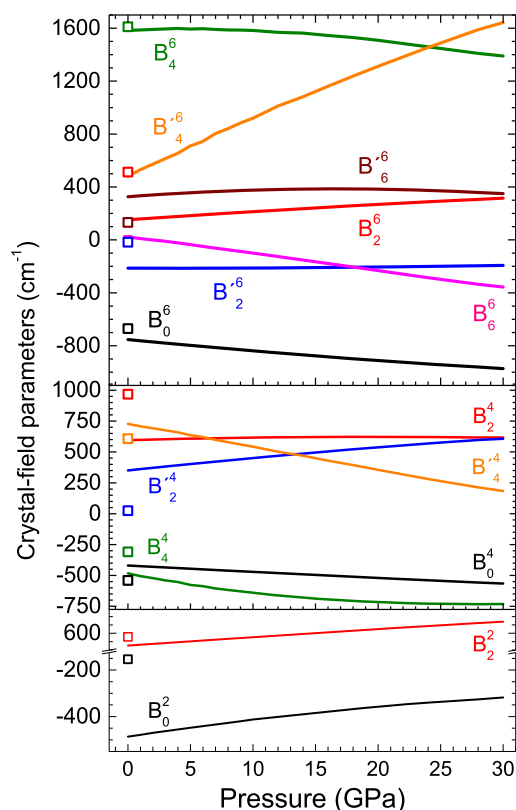


FIG. 6. Nd^{3+} crystal-field parameters calculated by the Simple Overlap Model (SOM) (solid lines) in YAlO_3 nano-perovskite. Experimental Stark levels by Duan *et al.*³⁰ at ambient conditions (open symbols) are also given.

high sensitivity to changes in relative angular positions (see [supplementary material](#)).

Simulations also confirm that the crystal-field interaction increases in magnitude as the pressure (volume) increases (decreases) when the CF strength parameter S (see [supplementary material](#)) is calculated as a function of pressure. Figure 7 shows an almost linear increase, from $\sim 550 \text{ cm}^{-1}$

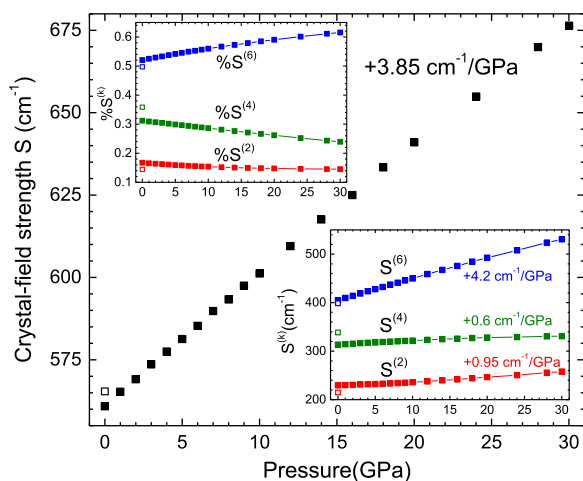


FIG. 7. Crystal-field strength parameter S as a function of pressure. Insets show the 2-, 4-, and 6-rank CF strength parameters $S^{(k)}$ and the contribution in percentage values $\%S^{(k)}$ as a function of pressure. Linear rates with pressure are shown for the $S^{(k)}$ parameters. Calculated values obtained from crystal field calculations by Duan *et al.*³⁰ at ambient conditions are also given (open symbols).

to $\sim 675 \text{ cm}^{-1}$, of the magnitude of the crystal-field interaction with pressure, as it is generally observed in many systems. Its relative change in YAlO_3 is similar to that found in $\text{MgO}:\text{LiNbO}_3$ oxide³⁷ but can be considered small compared to other oxide systems such as YVO_4 ³⁸ or $\text{Ca}_3\text{Ga}_2\text{Ge}_3\text{O}_{12}$ garnet.¹⁵ Increase of the total CF strength felt by the Nd^{3+} ions can be easily related to the shortening of Nd–O distances induced by pressure (see Fig. 2), which increases the Coulomb interaction between the f -electrons of the optically active ion and the valence electrons of oxygen ligands. This small change in the relative overall CF strength value has been previously related to high rigidity in the local (nearest) environment of the lanthanide, playing the role of an impurity.³⁷

However, for the discussion, it is better to analyse the contribution to the strength parameter of those CF parameters of the same k -rank, $S^{(k)}$, and their percentage contribution, $\%S^{(k)}$, to the total CF strength. The bottom inset in Fig. 7 shows how all CF parameters increase in magnitude giving rise to increasing $S^{(k)}$ contribution although 4-rank CF parameters show the lowest rate, while the 6-rank ones show the highest rate due to the large variations of B_4^6 and B_6^6 CF parameters. In the same way, $S^{(2)}$ contribution increases although it must be overestimated since only small changes are observed for the splitting of the $^4F_{3/2}$ emitting level, and its percentage contribution to the total strength is very small. The top inset in Fig. 7 shows how $\%S^{(2)}$ contributes in only 15% to the total CF value and this value remains almost unchanged up to $\sim 30 \text{ GPa}$, whereas $\%S^{(4)}$ percentage is reduced from 32% to 24%. On the contrary, $\%S^{(6)}$ parameter shows the largest pressure variation and the largest contribution to S values. Thus, as pressure increases, the influence of 6-rank parameters increases at expenses of those of 4-rank indicating that, since the dependence of crystal field B_q^k parameters with the distance from the central Nd^{3+} ion goes as $\sim (1/R^{k+1})$, the interaction between the Nd^{3+} ion and its nearest ligands is more intense at distances closer to the central lanthanide optically active ion, and the immediate environment contribution to the S value dominates, the more the higher the pressure. This means that the local environment surrounding the optically active ions is rigid enough to consider that very small displacements are allowed for the 8 nearest oxygen ligands and larger displacements for those 4 distant ones. This behavior would explain all the effects related to the free-ion and crystal-field interactions under pressure.

High pressure simulations show that all the barycenters of the $^{2S+1}L_J$ multiplets suffer wavelength blue-shifts, but with different rates. Looking closely at Fig. 4 one could see how, when the barycenter of the $^4F_{3/2}$ emitting multiplet has been taken as the origin of energies, the relative positions of the barycenters corresponding to both the $^4I_{9/2}$ and $^4I_{11/2}$ multiplets become closer as the pressure increases. This fact is due to the differences in their respective slopes of $+1.7 \text{ cm}^{-1}/\text{GPa}$, resulting in an effective approach between the emitting and low lying multiplets with pressure that can be related to the nephelauxetic effect. Conversely, the increase of the crystal-field interaction with pressure enlarges the splitting of both $^4F_{3/2}$ and $^4I_{9/2}$ states with different magnitudes (see [supplementary material](#)): while the splitting of the emitting $^4F_{3/2}$ doublet shows no sensitivity with pressure, the maximum splitting of the $^4I_{9/2}$

ground state is around $\sim 200\text{ cm}^{-1}$, with negative relative slopes for the Z_{1-3} Stark levels, and $\sim 106\text{ cm}^{-1}$ for the ${}^4I_{11/2}$ first excited state, with a negative relative slope for the Y_1 Stark level (see Fig. 4).

The combination of both effects as pressure increases, effective approaching of multiplets and increasing hyperfine splitting, explains the behaviors of the Stark energy level scheme and the emission lines observed in Figs. 3 and 4. Especially, they explain the overall blue-shifts (towards higher energies) of some emission lines associated with the ${}^4F_{3/2}(R_{1,2}) \rightarrow {}^4I_{9/2}(Z_{1-3})$ and ${}^4F_{3/2}(R_2) \rightarrow {}^4I_{11/2}(Y_1)$ transitions. Good agreement between calculated and experimental energies is found, with only a relatively large difference of around 50 cm^{-1} for the Z_4 Stark level, and although they are a slightly larger for the simulated emission line, it is worth noting how their tendencies with pressure are well described. Thus, typical behaviors, i.e., decrease of atomic parameters and increase of CF strength with pressure, are also observed for the Nd^{3+} optical properties in YAlO_3 nano-crystals.

IV. CONCLUSIONS

Pressure-induced energy blue- and red-shifts of the ${}^4F_{3/2} \rightarrow {}^4I_J$ ($J = 9/2, 11/2$) near-infrared emission lines of Nd^{3+} ions in YAlO_3 nano-crystalline perovskites have been observed under pressure. Although their pressure coefficient rates are smaller than those of ruby, they are large enough to consider $\text{YAlO}_3:\text{Nd}^{3+}$ as a candidate for the P-sensor in high-pressure experiments. These energy shifts have been related to pressure-induced changes in the interactions between the Nd^{3+} ions, located at the yttrium site, and their twelve oxygen ligands. These shifts have been explained using a combination of SOM simulations and *ab initio* calculations, using only the lanthanide ion first coordination environment, and have allowed us to successfully reproduce the trends observed in the experimental emission lines and the Stark energy level scheme at high pressures. Only a large difference, of around 50 cm^{-1} , between calculated and experimental energies has been found for the Z_4 Stark level although these differences are slightly larger for the simulated emission lines, their tendencies with pressure are well described. In addition, it is worth noting how these simulations have also allowed analyzing the main affects responsible for the optical properties of Nd^{3+} ions in YAlO_3 under pressure: a general increase of CF magnitude and a blue-shift of the average energy of the $4f^3$ configuration.

SUPPLEMENTARY MATERIAL

See [supplementary material](#) for the theoretical background of the simulation as well as the procedures followed for the fittings. Different tables related to Rietveld fittings, identification of the Nd^{3+} emission lines, and the experimental and calculated Nd^{3+} energy levels for different simulations were also included.

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